Name:	Student ID:

LATENT VARIABLES, MATRIX FACTORISATION AND MCMC PROBLEM SHEET

This sheet covers the lectures on probabilistic inference, matrix factorisation and MCMC. Question 1 practises generative and hierarchical models and the EM algorithm, Question 2 low-rank approximation and matrix completion, Question 3 drawing random samples, and Question 4 Markov chains and the Metropolis algorithm. Each ends with a short Python part (`import numpy as np` is assumed). Question 5 is harder and optional: it proves that EM and MCMC work, and runs alternating least squares.

1

(a) An email is spam with probability 0.4. The word "free" appears in 30% of spam emails and in 2% of other emails. Write down the joint distribution of S (spam or not) and F (contains "free" or not) as a 2×2 table. Find $\mathbb{P}(F = \text{yes})$ and $\mathbb{P}(S = \text{spam} \mid F = \text{yes})$. [3 marks]

$\overline{3}$

(b) In the hierarchical ratings model of the lecture, $\theta_j \sim \mathrm{Beta}(a,b)$ and $k_j \sim \mathrm{Bin}(n_j,\theta_j)$, the posterior mean of θ_j is $(a + k_j)/(a + b + n_j)$. Take $a = 20$, $b = 5$. Product C has 9 positive reviews out of 10, and product D has 85 out of 100. Rank them by the fraction of positive reviews and by the posterior mean. How many of its 10 reviews would C need to be positive to rank above D? [4 marks]

$\overline{4}$

(c) A latent variable Z is 1 or 0 with probability $\frac{1}{2}$ each. Given $Z = 1$, $X \sim \mathcal{N}(0,1)$; given $Z = 0$, $X \sim \mathcal{N}(4,1)$. Write down the density $f(x)$ of X. Show that the *responsibility* $\gamma(x) = \mathbb{P}(Z = 1 \mid X = x)$ is $1/(1 + \mathrm{e}^{4x-8})$, and evaluate it at $x = 1$, 2 and 3. [3 marks]

3

(d) Take the model of part (c) but treat $p = \mathbb{P}(Z = 1)$, μ_A and μ_B as unknown (the variances stay 1). The data are $X = (-1,0,1,2,4,5)$. Starting from $p = \frac{1}{2}$, $\mu_A = 0$, $\mu_B = 4$, carry out one step of EM: compute the responsibilities γ_i (use part (c)), then the new values

$$p = \frac{1}{m}\sum_{i=1}^{m}\gamma_i, \qquad \mu_A = \frac{\sum_i \gamma_i X_i}{\sum_i \gamma_i}, \qquad \mu_B = \frac{\sum_i (1-\gamma_i) X_i}{\sum_i (1-\gamma_i)}.$$

[4 marks]

4

(e) In the M-step the terms of Q that involve p are $\sum_{i=1}^{m}\big(\gamma_i \log(p)+(1-\gamma_i)\log(1-p)\big)$. Show that they are maximised by $p = \frac{1}{m}\sum_i \gamma_i$. [2 marks]

$\overline{2}$

(f) In Python, generate $m = 200$ points from the model with $p = 0.3$, $\mu_A = 0$, $\mu_B = 4$, start EM from $p = 0.5$, $\mu_A = -1$, $\mu_B = 1$, and print the log-likelihood $\sum_i \log\big(p\mathcal{N}(X_i \mid \mu_A,1) + (1-p)\mathcal{N}(X_i \mid \mu_B,1)\big)$ at each iteration. Does it ever decrease? [4 marks]

$\overline{4}$

End of question 1

(a) $\overline{3}$	(b) $\overline{4}$	(c) $\overline{3}$	(d) $\overline{4}$	(e) $\overline{2}$	(f) $\overline{4}$	Total $\overline{20}$

2

(a) A film site has 1000 users and 500 films. How many numbers does a rank $K = 20$ factorisation **AB** of the ratings matrix have, compared with the full matrix? LoRA adds a rank $K = 4$ update to a 1024×1024 weight matrix: what fraction of the weights does it train? [3 marks]

$\boxed{3}$

(b) For the ratings in the lecture

Name	Titanic	Dr No	The Matrix	Spartacus	Ten
Alice	2	Ø	4	5	2
Bob	5	3	Ø	3	Ø
Chas	Ø	2	3	5	3

use $R_{ui} \approx \bar{R}_u + \bar{R}_i - \bar{R}$ to predict Bob's rating of The Matrix and Chas's rating of Titanic. [4 marks]

$\boxed{4}$

(c) The matrix $\mathbf{X} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ is symmetric with positive eigenvalues, so its SVD is its eigen-decomposition. Find it, write down the best rank-1 approximation $\mathbf{X}_1$, and check that $\|\mathbf{X} - \mathbf{X}_1\|_F$ equals the discarded singular value, as Eckart–Young says. [4 marks]

$\overline{4}$

(d) In alternating least squares with $K = 1$ every factor is a number. A user has rated three films whose factors are $b_j = 1$, 2 and 0.5, giving ratings $X_j = 2$, 4 and 1. With the item factors fixed, the user's factor is the ridge regression solution $a = \sum_j b_j X_j / \left(\eta + \sum_j b_j^2\right)$. Find a and the predicted rating of a fourth film with $b = 1.5$ for $\eta = 0$, 0.5 and 5. Comment. [4 marks]

$\overline{4}$

(e) In Python, make an 8×6 matrix of rank 2, hide about a quarter of its entries, and fill them in with the iterative SVD method of the lecture (start from the user means; keep the best rank-2 approximation; copy its values into the missing entries only). Print the RMS error on the missing entries as you go. [5 marks]

5

End of question 2

(a) $\frac{}{3}$ (b) $\frac{}{4}$ (c) $\frac{}{4}$ (d) $\frac{}{4}$ (e) $\frac{}{5}$ Total $\frac{}{20}$

3

(a) We estimate $I = \int_0^1 \mathrm{e}^x \,\mathrm{d}x = \mathbb{E}\big[\mathrm{e}^U\big]$, with U uniform on $(0,1)$, by averaging e^{U_i} over n independent samples. Find $\mathrm{Var}(\mathrm{e}^U)$, the standard error for $n = 10^4$, and the n needed for a standard error of 10^{-4}. [4 marks]

$\overline{4}$

(b) The Cauchy density is $f(x) = 1/\big(\pi(1+x^2)\big)$. Find its cumulative distribution function and hence a transformation that turns a uniform deviate U into a Cauchy deviate. Why would averaging Cauchy samples be a bad way to estimate its mean? [4 marks]

$\overline{4}$

(c) We want samples from $f(x) = 6x(1-x)$ on $[0,1]$ using uniform proposals, $g(y) = 1$. Find the smallest c with $cg(x) \geq f(x)$, describe the rejection method, and find the fraction of proposals that are accepted. Explain why the accepted values have density f. [5 marks]

5

(d) To draw samples from $\mathcal{N}(\mathbf{0},\mathbf{I})$ in d dimensions we use proposals from the wider $\mathcal{N}(\mathbf{0},s^2\mathbf{I})$ with $s > 1$. Show that the smallest c is s^d. What fraction of proposals is accepted for $s = 1.2$ when $d = 1$, 10 and 100? [3 marks]

3

(e) In Python, implement the rejection sampler of part (c) with 10^5 proposals. Check the acceptance rate, and that the accepted samples have mean $\frac{1}{2}$ and variance $\frac{1}{20}$. [4 marks]

4

End of question 3

(a) $\frac{}{4}$	(b) $\frac{}{4}$	(c) $\frac{}{5}$	(d) $\frac{}{3}$	(e) $\frac{}{4}$	Total $\frac{}{20}$

4

(a) M_{ij} is the probability that a Markov chain moves from state j to state i. Explain why $\sum_i M_{ij} = 1$. Show that if $\boldsymbol{\pi}$ satisfies detailed balance, $M_{ij}\pi_j = M_{ji}\pi_i$, then $\mathbf{M}\boldsymbol{\pi} = \boldsymbol{\pi}$. What does this mean? [4 marks]

$\overline{4}$

(b) Four states sit on a ring, 1–2–3–4–1, with target probabilities $\boldsymbol{\pi} = (0.1, 0.2, 0.3, 0.4)$. From each state the Metropolis algorithm proposes one of its two neighbours, each with probability $\frac{1}{2}$, and accepts with probability $\min(1, \pi_i/\pi_j)$. Write down $\mathbf{M}$ and check that $\mathbf{M}\boldsymbol{\pi} = \boldsymbol{\pi}$. [6 marks]

$\overline{6}$

(c) In Python, find the eigenvalues of **M**, and starting from state 1, $\boldsymbol{p}(0) = (1,0,0,0)^{\mathsf{T}}$, print $\boldsymbol{p}(t) = \mathbf{M}^t\boldsymbol{p}(0)$ for t up to 10. How quickly does it approach $\boldsymbol{\pi}$? [3 marks]

3

(d) To sample from a posterior $\pi(\boldsymbol{\theta}) = f(\mathcal{D}|\boldsymbol{\theta})f(\boldsymbol{\theta})/f(\mathcal{D})$ the Metropolis algorithm accepts a proposal $\boldsymbol{\theta}'$ with probability $\min(1,\pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta}))$. Why don't we need $f(\mathcal{D})$? Why do programs compute $\log\pi(\boldsymbol{\theta}') - \log\pi(\boldsymbol{\theta})$ rather than the ratio itself? [3 marks]

3

(e) In Python, use Metropolis with proposals $x' = x + s\epsilon$, $\epsilon \sim \mathcal{N}(0,1)$, to sample from $\pi(x) \propto \mathrm{e}^{-(x^2-4)^2/8}$, which has peaks at $x = \pm 2$. Start at $x = 2$ and run 10^5 steps for $s = 0.1$, 1 and 10. For each, print the acceptance rate, the fraction of samples with $x > 0$ (which should be $\frac{1}{2}$), the number of times x changes sign, and the estimate of $\mathbb{E}\left[x^2\right]$ (whose exact value is 3.41). Comment. [4 marks]

$\frac{}{4}$

End of question 4

(a) $\frac{}{4}$	(b) $\frac{}{6}$	(c) $\frac{}{3}$	(d) $\frac{}{3}$	(e) $\frac{}{4}$	Total $\frac{}{20}$

5 This question is harder and optional. It shows why EM never decreases the likelihood, why an MCMC chain with detailed balance settles down, and runs alternating least squares on the lecture's ratings.

(a) For data $\mathcal{D}$, latent variables $\boldsymbol{Z}$ and parameters $\boldsymbol{\Theta}$, show that for every $\boldsymbol{Z}$

$$\log f(\mathcal{D}|\boldsymbol{\Theta}) = \log f(\mathcal{D}, \boldsymbol{Z}|\boldsymbol{\Theta}) - \log \mathbb{P}(\boldsymbol{Z}|\mathcal{D}, \boldsymbol{\Theta}).$$

Average both sides over $P_t(\boldsymbol{Z}) = \mathbb{P}\big(\boldsymbol{Z}|\mathcal{D}, \boldsymbol{\Theta}^{(t)}\big)$ to get $\log f(\mathcal{D}|\boldsymbol{\Theta}) = Q(\boldsymbol{\Theta}|\boldsymbol{\Theta}^{(t)}) + S(\boldsymbol{\Theta})$, where Q is the function EM maximises and $S(\boldsymbol{\Theta}) = -\sum_{\boldsymbol{Z}} P_t(\boldsymbol{Z}) \log \mathbb{P}(\boldsymbol{Z}|\mathcal{D}, \boldsymbol{\Theta})$. Using $\log(x) \leq x - 1$, show that $S(\boldsymbol{\Theta}) \geq S(\boldsymbol{\Theta}^{(t)})$ for every $\boldsymbol{\Theta}$, and hence that an EM step cannot decrease the likelihood.

[7 marks]

7

(b) Let $\mathbf{D} = \mathrm{diag}(\boldsymbol{\pi})$ with all $\pi_i > 0$. Show that if **M** satisfies detailed balance then $\mathbf{A} = \mathbf{D}^{-1/2}\mathbf{M}\mathbf{D}^{1/2}$ is symmetric. Deduce that the eigenvalues of **M** are real and that **M** has a full set of eigenvectors (the assumption made in the lecture when we expanded $\boldsymbol{p}(0)$ in eigenvectors). Check that $A_{12} = A_{21}$ for the matrix of Question 4(b). [6 marks]

6

(c) In matrix factorisation with **B** fixed, row $\boldsymbol{a}_i$ of **A** minimises $\sum_j (X_{ij} - \boldsymbol{a}_i^{\mathsf{T}}\boldsymbol{b}_j)^2 + \eta\|\boldsymbol{a}_i\|^2$, the sum running over the films user i has rated. Differentiate to show that $\boldsymbol{a}_i = \left(\eta\mathbf{I} + \sum_j \boldsymbol{b}_j\boldsymbol{b}_j^{\mathsf{T}}\right)^{-1}\sum_j \boldsymbol{b}_j X_{ij}$, and check that it reduces to the formula of Question 2(d) when $K = 1$. Why is the matrix being inverted always invertible when $\eta > 0$? [5 marks]

5

(d) In Python, run alternating least squares with $K = 1$ and $\eta = 0.1$ on the ratings of Question 2(b) (use `np.nan` for the missing entries), starting from random factors. Compare the predictions for Bob's rating of The Matrix and Chas's rating of Titanic with your answers to Question 2(b). [7 marks]

$\overline{7}$

End of question 5

(a) $\overline{7}$	(b) $\overline{6}$	(c) $\overline{5}$	(d) $\overline{7}$	Total $\overline{25}$